

An Analysis of Machine translation

Machine translation

Machine translation -- Part 1

Transfer-based machine translation was similar to interlingual machine translation in that it created a translation from an intermediate representation that simulated the meaning of the original sentence. Unlike interlingual MT, it depended partially on the language pair involved in the translation.

Machine translation -- Part 2

International Association for Machine Translation (IAMT) Archived 24 June 2010 at the Wayback Machine Machine Translation Archive Archived 1 April 2019 at the Wayback Machine by John Hutchins. An electronic repository (and bibliography) of articles, books and papers in the field of machine translation and computer-based translation technology Machine translation (computer-based translation) – Publications by John Hutchins (includes PDFs of several books on machine translation) Machine Translation and Minority Languages John Hutchins 1999 Archived 7 September 2007 at the Wayback Machine

Machine translation -- Part 3

Surveillance and military Following terrorist attacks in Western countries, including 9-11, the U.S. and its allies have been most interested in developing Arabic machine translation programs, but also in translating Pashto and Dari languages. Within these languages, the focus is on key phrases and quick communication between military members and civilians through the use of mobile phone apps. The Information Processing Technology Office in DARPA hosted programs like TIDES and Babylon translator. US Air Force has awarded a \$1 million contract to develop a language translation technology.

Machine translation -- Part 4

Public administration Despite their inherent limitations, MT programs are used around the world. Probably the largest institutional user is the European Commission. In 2012, with an aim to replace a rule-based MT by newer, statistical-based MT@EC, The European Commission contributed 3.072 million euros (via its ISA programme).

Machine translation -- Part 5

Applications While no system provides the ideal of fully automatic high-quality machine translation of unrestricted text, many fully automated systems produce reasonable output. The quality of machine translation is substantially improved if the domain is restricted and controlled. This enables using machine translation as a tool to speed up and simplify translations, as well

as producing flawed but useful low-cost or ad-hoc translations.

Machine translation -- Part 6

Interlingual machine translation was one instance of rule-based machine-translation approaches. In this approach, the source language, i.e. the text to be translated, was transformed into an interlingual language, i.e. a "language neutral" representation that is independent of any language. The target language was then generated out of the interlingua. The only interlingual machine translation system that was made operational at the commercial level was the KANT system (Nyberg and Mitamura, 1992), which was designed to translate Caterpillar Technical English (CTE) into other languages.

Machine translation -- Part 7

A deep learning-based approach to MT, neural machine translation has made rapid progress in recent years. However, the current consensus is that the so-called human parity achieved is not real, being based wholly on limited domains, language pairs, and certain test benchmarks, i.e., it lacks statistical significance power. Translations by neural MT tools like DeepL Translator, which is thought to usually deliver the best machine translation results as of 2022, typically still need post-editing by a human. Instead of training specialized translation models on parallel datasets, one can also directly prompt generative large language models like GPT to translate a text. This approach is considered promising, but is still more resource-intensive than specialized translation models.

Machine translation -- Part 8

Non-standard speech One of the major pitfalls of MT is its inability to translate non-standard language with the same accuracy as standard language. Heuristic or statistical based MT takes input from various sources in standard form of a language. Rule-based translation, by nature, does not include common non-standard usages. This causes errors in translation from a vernacular source or into colloquial language. Limitations on translation from casual speech present issues in the use of machine translation in mobile devices.

Machine translation -- Part 9

1990s and early 2000s By 1998, "for as little as \$29.95" one could "buy a program for translating in one direction between English and a major European language of your choice" to run on a PC. MT on the web started with SYSTRAN offering free translation of small texts (1996) and then providing this via AltaVista Babelfish, which racked up 500,000 requests a day

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(1997). The second free translation service on the web was Lernout & Hauspie's Globalink. Atlantic Magazine wrote in 1998 that "Systran's Babelfish and Globalink's Comprendre" handled "Don't bank on it" with a "competent performance." Franz Josef Och (the future head of Translation Development AT Google) won DARPA's speed MT competition (2003). More innovations during this time included MOSES, the open-source statistical MT engine (2007), a text/SMS translation service for mobiles in Japan (2008), and a mobile phone with built-in speech-to-speech translation functionality for English, Japanese and Chinese (2009). In 2012, Google announced that Google Translate translates roughly enough text to fill 1 million books in one day.

Machine translation -- Part 10

Word-sense disambiguation concerns finding a suitable translation when a word can have more than one meaning. The problem was first raised in the 1950s by Yehoshua Bar-Hillel. He pointed out that without a "universal encyclopedia", a machine would never be able to distinguish between the two meanings of a word. Today there are numerous approaches designed to overcome this problem. They can be approximately divided into "shallow" approaches and "deep" approaches. Shallow approaches assume no knowledge of the text. They simply apply statistical methods to the words surrounding the ambiguous word. Deep approaches presume a comprehensive knowledge of the word. So far, shallow approaches have been more successful. Claude Piron, a long-time translator for the United Nations and the World Health Organization, wrote that machine translation, at its best, automates the easier part of a translator's job; the harder and more time-consuming part usually involves doing extensive research to resolve ambiguities in the source text, which the grammatical and lexical exigencies of the target language require to be resolved:

Machine translation -- Part 11

There are many factors that affect how machine translation systems are evaluated. These factors include the intended use of the translation, the nature of the machine translation software, and the nature of the translation process. Different programs may work well for different purposes. For example, statistical machine translation (SMT) typically outperforms example-based machine translation (EBMT), but researchers found that when evaluating English to French translation, EBMT performs better. The same concept applies for technical documents, which can be more easily translated by SMT because of their formal language. In certain applications, however, e.g., product descriptions written in a controlled language, a dictionary-based

machine-translation system has produced satisfactory translations that require no human intervention save for quality inspection. There are various means for evaluating the output quality of machine translation systems. The oldest is the use of human judges to assess a translation's quality. Even though human evaluation is time-consuming, it is still the most reliable method to compare different systems such as rule-based and statistical systems. Automated means of evaluation include BLEU, NIST, METEOR, and LEPOR. Relying exclusively on unedited machine translation ignores the fact that communication in human language is context-embedded and that it takes a person to comprehend the context of the original text with a reasonable degree of probability. It is certainly true that even purely human-generated translations are prone to error. Therefore, to ensure that a machine-generated translation will be useful to a human being and that publishable-quality translation is achieved, such translations must be reviewed and edited by a human. The late Claude Piron wrote that machine translation, at its best, automates the easier part of a translator's job; the harder and more time-consuming part usually involves doing extensive research to resolve ambiguities in the source text, which the grammatical and lexical exigencies of the target language require to be resolved. Such research is a necessary prelude to the pre-editing necessary in order to provide input for machine-translation software such that the output will not be meaningless. In addition to disambiguation problems, decreased accuracy can occur due to varying levels of training data for machine translating programs. Both example-based and statistical machine translation rely on a vast array of real example sentences as a base for translation, and when too many or too few sentences are analyzed accuracy is jeopardized. Researchers found that when a program is trained on 203,529 sentence pairings, accuracy actually decreases. The optimal level of training data seems to be just over 100,000 sentences, possibly because as training data increases, the number of possible sentences increases, making it harder to find an exact translation match. Flaws in machine translation have been noted for their entertainment value. Two videos uploaded to YouTube in April 2017 involve two Japanese hiragana characters ■■ (e and gu) being repeatedly pasted into Google Translate, with the resulting translations quickly degrading into nonsensical phrases such as "DECEARING EGG" and "Deep-sea squeeze trees", which are then read in increasingly absurd voices; the full-length version of the video currently has 7.1 million views as of August 2025.